

# The Plan-EO Pipeline: Supporting Enteric Disease Modeling with Structured, Scalable Data Infrastructure

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## Abstract

Diarrheal diseases caused by ecologically sensitive pathogens remain a leading cause of childhood illness, particularly in low- and middle-income countries (LMICs). Despite increasing awareness of the links between environmental conditions and human health, enteric diseases have not received a proportionate amount of attention in global health research [1]. The Planetary Child Health and Enterics Observatory (Plan-EO) is an interdisciplinary initiative designed to address this gap by producing spatial data products that map pathogen distribution alongside environmental and sociodemographic variables. To support Plan-EO’s research goals, our data science capstone team has built a scalable data pipeline that integrates diverse datasets, including survey, population, and geographic data, and prepares them for use in predictive models and interactive tools. By improving data accessibility and usability, our work strengthens the infrastructure necessary to inform and target child health interventions more effectively.

## 1 Introduction

Plan-EO aims to improve child health outcomes in low- and middle-income countries by generating research-driven insights into the environmental and sociodemographic factors that influence the transmission of enteric diseases [1]. The achievement of these goals depends heavily on the availability of accurate and accessible data. However, the management of heterogeneous datasets, including geographic information, population distributions, and international health surveys, poses logistical and technical challenges.

To address these challenges, our capstone team developed a robust and semi-automated data pipeline designed to streamline the col-

lection, integration, and organization of key datasets. The pipeline consolidates data from multiple sources and formats and integrates them into a standardized structure, improving data quality, enhancing reliability, and increasing operational efficiency. By strengthening the research infrastructure supporting Plan-EO, our work facilitates more rapid and effective evidence-based interventions aimed at reducing the global burden of enteric diseases in children.

## 2 Background

Recent work in enteric disease modeling has highlighted the importance of integrating

large-scale, multi-source datasets. The original Plan-EO architecture developed by Colston et al. (2024) [1], laid the foundation for combining geospatial, environmental, and health survey data. However, the initial implementation relied heavily on manual data collection, which limited efficiency and scalability. Since data pipeline creation and management takes a considerable amount of time and effort, projects typically appoint a dedicated person/team to this task [2]. Our capstone team assumed that role for Plan-EO.

Beyond initially lacking a formal pipeline, the original data collection also faced obstacles with automation, reconciling inconsistent metadata, and managing updates, challenges not unique to Plan-EO. As described by Munappy et al. (2025) [2], pipeline development frequently encounters issues related to infrastructure fragility, documentation gaps, and data quality. These problems can be amplified in global health contexts, where data often originate from decentralized surveys (e.g., DHS MICS) or open source mapping platforms (e.g., OSM), making standardization and reproducibility increasingly difficult.

We designed our pipeline in response to these challenges, drawing on best practices in data engineering and public health informatics. Core ideas such as modularity, traceability, and metadata fidelity, which are widely recognized as foundational in modern data pipelines, were important to our design [2, 3]. Organizations like the Institute for Health Metrics and Evaluation (IHME) emphasize transparency and documentation for managing large health datasets [4], while the World Health Organization’s (WHO) *Health Data Toolkit* promotes national health data integration through standardization practices [5]. Drawing on these frameworks, the Plan-EO pipeline incorporates structured logging and extensive documentation, including user guides and reference metadata, to improve usability and align with global standards for

interoperability and standardization.

At its core, a data pipeline is code that removes manual steps from data collection workflows and allows for a streamlined and automated flow of information from one repository to another [2]. That is what we worked to create for Plan-EO. Our work focused not only on the technical structure but also ensured the pipeline would be generalizable and usable in the future. Although we developed a pipeline more tailored to the Plan-EO project, we recognize an opportunity for our work to be used as a more generalized framework, one that could serve as a blueprint for other researchers and practitioners working with large, multi-source datasets, particularly in health, demographic, or geospatial contexts. By including robust documentation and logging mechanisms, we can ensure that future users, whether part of Plan-EO or other projects, can easily adopt and extend our data pipeline.

## 3 Data Description

The Plan-EO data pipeline integrated information from four data repositories: the Demographic and Health Surveys (DHS), the Multiple Indicator Cluster Surveys (MICS), Open Street Map (OSM), and WorldPop.

### 3.1 Demographic and Health Surveys (DHS)

The DHS Program provides nationally representative surveys that are designed to support monitoring and evaluation indicators important for both country-specific and cross-national comparisons [6]. To date, DHS has conducted more than 400 surveys across over 90 countries incorporating questionnaires, biomarker collection, and geospatial boundary data. Examples of indicators collected by DHS include main flooring material

used in housing and child vaccination status. Plan-EO draws on DHS datasets collected from 2003 to the present which are available in a variety of formats including Plan-EO researchers’ preferred Stata files. Due to restructuring, USAID, which runs DHS, has been defunded and no new DHS surveys will be produced.

### 3.2 Multiple Indicator Cluster Surveys (MICS)

The MICS program, administered by UNICEF, makes up the worlds largest repository of reliable and internationally comparable data on women and children [7]. Over 400 surveys have been conducted in more than 120 countries dating back to 1995, and they serve as an integral part of national health policies. MICS surveys cover health, education, and child protection information, including indicators like access to clean drinking water and percentage of children underweight. The MICS data is available as SPSS/.sav files.

### 3.3 OpenStreetMap (OSM)

OpenStreetMap is an open-source geographic data repository maintained by a community of volunteers. The database combines information from surveys, aerial photos, satellite imagery, and other sources [8]. For Plan-EO, we extracted point data for thousands of settlements, in some cases down to the individual dwelling level, and converted those points to both CSV and shapefile format.

### 3.4 WorldPop

WorldPop is a well-documented, open access repository that provides “high resolution, contemporary data on human population distributions and their compositions” [9]. Population density data are crucial for

child health interventions and form an integral part of our pipeline. We primarily used 1km-resolution GeoTIFF files, which offered an ideal balance between detail and memory constraints.

## 4 Methodology

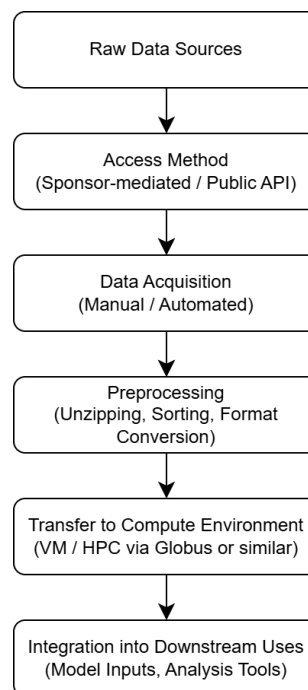


Figure 1: A generalized data pipeline architecture for multi-source health and geospatial datasets.

For all four data sources, we followed a standard data pipeline architecture (see Figure 1). This pipeline serves as a structured approach for data acquisition, preprocessing, and integration into downstream analyses. Although specific access methods and preprocessing steps vary across sources, the overall framework remains the same. Raw data is accessed and acquired, then preprocessed and transferred to a secure compute environment before being integrated into modeling workflows.

Since Plan-EO uses human health data, the team uses a HIPAA-compliant virtual machine (VM) for data storage and analysis. To maintain the security of the environment, it is isolated from direct internet access. Therefore, data cannot be downloaded directly to the VM, but instead, the initial data acquisition is done within a shared workspace on the University of Virginia’s Rivanna Research Computing environment. Data are then transferred to the VM via Globus, a high-security file transfer software.

The following sections outline the access methods, data acquisition, and preprocessing steps used for each data source.

## 4.1 Demographic and Health Surveys (DHS)

To collect a comprehensive set of DHS datasets, we adopted a hybrid strategy that combined two primary methods: direct download from stable URLs and web scraping. Because login credentials were required to access any survey data, we automated authentication using a Selenium-driven WebDriver. For survey data accessible via static and predictable URLs, we issued simple HTTP requests to download content directly, reducing both computational complexity and overhead when compared to scraping.

When stable download links were unavailable, particularly for geospatial boundary datasets, web scraping was necessary. Knowing the server strain web scraping can cause, we restricted this to a one-time collection. Due to the dynamic page structure of the DHS spatial data repository, traditional HTML parsing was ineffective, so we again employed `Selenium` to simulate user interactions and handle the complex page structure.

Although DHS provides an API, it is limited to metadata retrieval. However, we leveraged this API to supplement our data by integrating consistent metadata fields that

proved valuable in downstream processing.

After download, we parsed and categorized the data by country and survey year. Although much of this process was automated by extracting information from filenames and URLs, some cases required manual intervention and review due to missing or inconsistent metadata. We systematically logged these exceptions to ensure transparency and reproducibility.

## 4.2 Multiple Indicator Cluster Surveys (MICS)

Accessing MICS data posed unique challenges. API requests were rejected and scraping attempts were blocked, necessitating manual batch downloads organized by survey round. This limitation reduced automation at this stage and decreased overall pipeline efficiency.

MICS datasets exhibited neither standard naming convention nor regularized internal directory structures. Some zip files included only country names, while others inconsistently referenced territories, regions, or survey years. Misspellings, regional subdivisions, and non-standard country abbreviations further complicated our parsing efforts.

To address these inconsistencies, we implemented extensive conditional logic and flexible parsing strategies. Despite our efforts to build robust and generalizable code, complete automation was not feasible, and ongoing logging, implemented via Python’s `logger` module, is critical for long-term functionality and debugging. To deal with survey updates and facilitate future use by Plan-EO researchers, we developed a Python module with detailed instructions that enables streamlined, reproducible preprocessing of MICS datasets.

## 4.3 OpenStreetMap (OSM)

The OpenStreetMap settlement data is the most frequently updated of any data we

worked with. We used the main public instance of Overpass API, a read-only API for querying OSM’s database. Initialization was the most challenging process: retrieving all points of interest for every Plan-EO country. With over 3 million initial data points, rate-limiting presented a serious issue. It took several hours, with significant cooldown periods between API calls, to download all of the necessary data using Python’s `requests` library. This was an important hurdle because it pushed us to consider handling rate-limiting within the pipeline.

For each country, the returned JSON file was first organized into a `pandas` dataframe and written to the output directory as a CSV. Using `geopandas`, we then converted all settlements with associated latitude and longitude (the majority for most countries) into a spatial dataframe, which was then outputted as a shapefile.

To automate the update of these files, we wrote a new function to retrieve the previous edit timestamp of each file and limit our Overpass query to settlements that had been edited in OSM more recently. This minimized storage and network strain and made the update process much faster than initialization, averaging around 30 minutes. We also introduced error handling for countries with no recent updates and error handling for a query timeout, which triggers a five-minute automatic cooldown period. When the pipeline is in use, errors are accounted for without requiring any technical expertise on the part of the user. With tens of thousands of updates per month, this will be a significant improvement over manual downloads.

## 4.4 WorldPop

The WorldPop data was the most straightforward to collect. The open-access WorldPop population data API exposed stable download links for GeoTIFF and ASCII files

under a variety of different aliases. Each of these aliases represented a different series of WorldPop data, such as "2000-2020 Unconstrained 1km resolution UN-adjusted". Using Python’s `requests` and `urlretrieve` libraries, we downloaded and organized the most recent 1km-resolution population density GeoTIFFs for each of the Plan-EO countries. Although updates to this data are less frequent, we documented our work to make it easy to update the workflow when more series become available. We assume that the 2015-2030 series will eventually be released at the 1km resolution.

Though our primary purpose in developing this pipeline was providing the most up-to-date data available, resource management limited our ability to do so with WorldPop. While there were GeoTIFFs available for population density for the years 2020-2024 at the 100m resolution, as part of the 2015-2030 series, the most recent available data at the 1km resolution was only from 2020. In collaboration with our sponsor, we determined that using the 100m resolution would result in a relatively negligible benefit to the modeling process, and each file would be 100 times larger, with corresponding strains on storage, network resources, and model training. Therefore, we chose not to collect the most recent data and our warehouse currently only contains WorldPop information up to 2020.

## 5 Results

The Plan-EO pipeline successfully automated the majority of the data collection and organization tasks. We collected nearly 3000 DHS datasets from 400 surveys spanning from 1985 to present along with corresponding geographic boundary data from the DHS Spatial Repository covering over 90 countries. In addition, we retrieved 265 MICS datasets

that span a comparable temporal range. We compiled 3.5 million OSM settlement data points and population density data for every LMIC targeted by the Plan-EO Initiative.

Approximately 80 percent of our data acquisition was automated through a combination of API calls, web scraping, and stable download links. Metadata extraction was successful for the majority of survey datasets; however, a small portion required manual metadata retrieval due to inconsistencies and/or missing documentation. Flexible parsing strategies were deployed to accommodate variations in file naming and archive structures, minimizing the need for manual intervention.

Although OSM data initially faced challenges due to API rate limits and large data volume, these issues were successfully mitigated. Future OSM data collection is expected to be more efficient, as incremental updates will involve significantly less data.

The resulting data warehouse is organized by country labeled with a standardized country name and ISO3 code, with subdirectories for survey, geographic, and population data. This structure supports Plan-EO’s initiatives by allowing for efficient and scalable disease modeling.

## 6 Discussion

Our implementation of a semi-automated scalable data pipeline represents a significant advancement in supporting Plan-EO’s modeling efforts. By standardizing spatial, survey, and population data, our pipeline substantially reduces the time and effort required for data collection and preparation, thereby accelerating research workflows.

Beyond Plan-EO, our pipeline framework is more broadly applicable to global health and spatial epidemiology projects that face similar challenges of data fragmentation and

inconsistent metadata. Our methods for flexible data parsing, modular design, and metadata extraction can reduce common barriers to large-scale international studies that have historically been limited by data access challenges.

Despite the pipeline’s benefits, several challenges remain. Although semi-automation greatly reduced manual data processing, edge cases involving missing metadata or nonstandard file structure still required human intervention. Moreover, a fundamental limitation lies in our pipelines reliance on stable upstream data formats, introducing the need for long-term maintenance to ensure continued functionality.

## 7 Conclusion

We have successfully developed a scalable semi-automatic pipeline for acquiring and preprocessing large-scale public health data, thereby enabling more efficient modeling by Plan-EO researchers. The pipeline serves as a generalizable template for others working with decentralized data and reduces barriers to multi-country and multi-source data analysis.

To enhance the pipeline’s capabilities, future work could focus on real-time data update mechanisms such as scheduled API queries, automated data transfers, and alert systems for new data availability. Additionally, expanding the scope of supported datasets to include hydrological, climatological, and ecological variables could enhance the pipeline’s applicability across a broader range of global health domains. By continuing to build on this foundation, Plan-EO and similar projects can accelerate evidence-based child health interventions.

## 8 Acknowledgments

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## B Code Repository

The project repository can be accessed at: <https://github.com/rsfellman/Plan-EO>